MILITARY SPECIFICATION

MOUNT, SIGHT: 11727200

1. SCOPE

1.1 This specification covers one type of Mount known as Mount, Sight, 11727200, which supports a Crew-Served Weapon Night Vision Sight.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MIL-F-13926	Fire Control Materiel; General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Army Materiel; Packaging and Packing of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables; for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions

DRAWINGS

U.S. Army, Frankford Arsenal

F11727200	Mount, Sight: 11727200

PACKAGING DATA SHEET

11727200	Packaging of Mount, Sight: 11727200

(Copies of specifications, standards, drawing and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 1240

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3. REQUIREMENTS

3.1 Materials.- The materials shall be in accordance with the drawings, material specifications and general specifications forming a part of this specification.

3.2 Fabrication.- The body assembly shall be manufactured in accordance with Drawing F11727200 and drawings pertaining thereto.

3.3 General specifications.- The contractor shall be responsible for adherence to, and compliance with, the following requirements of Specification MIL-F-13926:

(a) Order of precedence
(b) Dimensions and tolerances
(c) Inorganic protective surface finishes
(d) Part identification and marking
(e) Workmanship

3.4 Environmental.

3.4.1 Storage temperature.- The mount assembly shall show no evidence of physical failure, and shall meet all the requirements of this specification at standard ambient temperature (plus 60° to 90°F) after having been exposed and thermally stabilized at ambient temperatures of minus 80° and plus 160° Fahrenheit.

3.4.2 Operating temperatures.- The mount assembly shall meet the requirements of 3.5.2 while exposed and thermally stabilized at ambient temperatures of minus 40° and plus 150°F, subsequent to which the mount assembly shall meet the requirements of 3.5.1, 3.5.3, 3.5.4 and 3.5.5 at the standard ambient temperature (plus 60° to 90°F).

3.4.3 Vibration.- The mount assembly shall be vibrated at a constant frequency of 30 cycles per second with an amplitude of 1/16 inch (1/8 inch total excursion) for a period of 5 minutes plus or minus 15 seconds. Subsequent to vibration, there shall be no evidence of physical failure and the mount assembly shall meet the requirements of 3.5.1 to 3.5.5 inclusive.

3.5 Performance.

3.5.1 Alignment.- With base mounting surface of the mount level, the dovetail key locating surface shall be parallel to the base mounting surface of the mount within .0005 inch when the centerline of the arm is at an angle of 39° ± 1'.

3.5.2 Travel.- After alignment has been established, rotation of the arm shall cause unrestricted movement of the mount through an angular movement from 15 degrees depression to 60 degrees elevation.

3.5.3 Plumb travel.- With a line of sight representing the dovetail key locating surface in coincidence with a plumb line at 60 degrees

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elevation the line of sight shall not deviate from the plumb line by more than 1.0 mil throughout the full range of travel.

3.5.4 Repeatability.- The holder, when open, shall permit unrestricted installation or removal of the sight. The holder shall permit repeated installation and removal of the sight without deviating the line of sight more than .15 mil after clamping.

3.5.5 Backlash.- Backlash at any position within the 75° range shall not exceed 0.1 mil.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article (initial production) approval.- The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of three (3) mount assemblies plus three (3) each of all items covered by SQAP. The sample shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing.- When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

4.2.2 Contractor testing.- When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

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4.3 Inspection provisions.

4.3.1 Submission of product.- Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies.- All components and subassemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality Assurance Provisions (SQAP) listed in the technical data package (TDP). In the absence of SQAP's, the applicable Quality Assurance Provisions of MIL-F-13926 shall apply. Examination, and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Tables I, II and III herein. Examination and tests for packaging, packing, and marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects in Tables I, II, and III shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

TABLE I - CLASSIFICATION OF DEFECTS

CLASS	REQUIREMENT	TEST PROCEDURE
CRITICAL: NONE		
MAJOR: AQL 1.0% DEFECTIVE		
101. Vibration	3.4.3	4.6.1.1
102. Alignment	3.5.1	4.6.2.1
103. Travel	3.5.2	4.6.2.2
104. Plumb travel	3.5.3	4.6.2.3
105. Backlash	3.5.5	4.6.2.4
106. Repeatability	3.5.4	4.6.2.5
MINOR: NONE		

NOTE: The tests in Table I shall be conducted at a temperature between 60° and 90°F.

4.3.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Special Sampling.

4.4.1 General.- One mount shall be selected at random as a special sampling from each 100 produced. The sample shall meet the requirements and tests in Table II.

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TABLE II

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
301.	Fabrication	3.2	Applicable Drawing Visual
302.	General specification	3.3	MIL-F-13926, Visual

4.4.2 Environmental.- Three (3) mounts shall be selected at random as samples from each 50 produced, or from each month's production, whichever occurs first. Each sample shall meet all the requirements and tests in Table III and shall then meet the requirements and tests in Table I.

TABLE III

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
303.	Storage temperature (-80° and +160°F)	3.4.1	4.6.1.2
304.	Operational temperature (Mechanical Operability -40°F)	3.4.2	4.6.1.3
305.	Operational temperature (Mechanical Operability +150°F)	3.4.2	4.6.1.3

4.4.3 Failure of sample.- Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections have been made by the contractor and resubmitted samples have been approved. (See 4.3.3)

4.5 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design.- Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications or SQAP forming a part of the contract shall be supplied by the contractor in accordance with technical data listed in the technical data package (TDP).

4.5.2.2 Contractor design.- The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of

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MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.6 Test methods and procedures.

4.6.1 Environmental.

4.6.1.1 Vibration.- To determine conformance to 3.4.3, the mount shall be mounted separately or collectively on a suitable mounting apparatus and attached to a vibrating machine or fixture (see 4.5.2.2) and vibrated at the amplitude, frequency, and time specified. After vibration, the units shall be given a visual-tactile examination for damage and loose parts and shaft movement. The units shall then be subjected to the tests specified in 4.6.2.1 to 4.6.2.5 inclusive to determine conformance to the respective requirements referenced therein.

4.6.1.2 Storage temperatures.- To determine conformance to 3.4.1, the mount shall be tested in accordance with Temperature Cycling Procedure II of MIL-F-13926 for the specified temperatures. At the conclusion of these tests, the units shall show no evidence of damage or deterioration when visually examined. Caution shall be exercised during this test to avoid subjecting the mount to thermal shock.

4.6.1.3 Operating temperatures.- To determine conformance to 3.4.2, the mount shall be placed in appropriate environmental chambers (see 4.5.2.2). The mount shall be exposed to each temperature extreme until thermally stabilized at the specified temperature in 3.4.2. While at each temperature extreme, the connector assembly shall be manually rotated as specified in 3.5.2. During this rotation, there shall be no binding or restriction of travel of the mount. Subsequent to the extreme temperature exposures, the mount shall be stabilized in an ambient temperature between plus 60° and 90°F and shall be subjected to the tests specified in 4.6.2.1, 4.6.2.3, 4.6.2.4 and 4.6.2.5 to determine conformance to the respective requirements referenced therein. Caution shall be exercised during this test to avoid subjecting the mount to thermal shock.

4.6.2 Performance tests.

4.6.2.1 Alignment test.- To determine conformance to 3.5.1, the mount shall be placed on a surface plate with the base mounting surface bearing on the plate. The trunnion arm shall be placed at the angle specified and the dovetail keyway shall be checked with an indicating height gage for parallelism.

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4.6.2.2 Travel test.- After alignment has been established (see 4.6.2.1), the trunnion shaft shall be actuated to a 15 degree depression and a 60 degree elevation, as measured with a sine bar to determine compliance with 3.5.2.

4.6.2.3 Plumb travel test.- To determine conformance to 3.5.3, the surface plate shall be levelled (see 4.6.2.1) and an alignment telescope shall be clamped to the dovetail key parallel to the dovetail. With the reticle of the scope on a plumb line, the test of 4.6.2.2 shall be repeated and the alignment scope reticle shall not deviate from the plumb line in excess of the tolerance specified in 3.5.3.

4.6.2.4 Backlash.- Conduct this test concurrent with the test in 4.6.2.3. In conjunction with the set-up in 4.6.2.3, the test for backlash shall be accomplished utilizing a precision 10 second bench level and a wall target. Means shall be provided to position the bench level with the alignment telescope parallel to the dovetail key. The target shall contain a plumb line with a calibrated horizontal zero elevation line capable of measuring backlash tolerance as specified in 3.5.5. Proceed with the test by observing through the alignment telescope. With backlash excluded for the intended direction of vertical travel, adjust the line of sight of the telescope in coincidence with the established zero elevation line on the target. At this setting, the bubble of the precision bench level shall be centered. By means of utilizing Connector Assembly 11727206, rotate the trunnion arm approximately 10 degrees in the same direction of travel, then reverse the direction of travel of the trunnion arm (with backlash included) and re-establish the zero elevation position by centering the level vial bubble in the precision bench level without overtravel. The deviation between the telescope horizontal reticle line and the target zero reticle line shall not be in excess of the backlash tolerance specified in 3.5.5. This test shall be repeated at +60° elevation and -15° depression by means of utilizing a sine bar with the precision bench level vial to center the level vial bubble at the respective limits.

4.6.2.5 Repeatability test.- To determine conformance to 3.5.4, an alignment telescope simulating the external dimensions of the sight shall be inserted in the dovetail keyway. The scope shall be aligned on a target then clamped and any change in alignment noted. The check shall be repeated three more times and the four alignment deviations averaged to determine conformance. Installation and removal of the simulated sight shall be as specified.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.- Packaging, packing and marking shall be in accordance with packaging data sheet 11727200. The level of protection shall be as specified in the procurement document (see 6.2b).

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6. NOTES

6.1 Intended use.- To mount Crew Served Weapon Night Vision Sight on Cupola M1, (Tank M48A3).

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Selection of applicable levels of preservation, packaging and packing.
(c) Applicable packaging data sheet number (see 5.1).
(d) Applicable stock number.
(e) Requirement for first article inspection.

Custodian:

Army - MU

Preparing activity:

Army - MU

Project No. 1240-A121

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Philadelphia, Pa. 19137

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

[ ] YES [ ] NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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